

Gene Sight Andro-Male Infertility NGS Panel

Patient name	: Mr. XX	PIN	: XX
Gender/Age	: Male / XX Years	Sample number	: XX
Referring Clinician	: XX	Sample collection date	: NA
Hospital/ Clinic	: XX	Sample received date	: NA
Specimen	: Peripheral blood	Report date	: NA

INDICATION FOR TESTING

Proband, Mr. XX is of non-consanguineous marriage. The couple has undergone 2 ART cycles. In the first cycle, self-gametes were used; 9 oocytes were retrieved, and fertilization failure was observed. In the second cycle, a donor oocyte was used; 10 oocytes were retrieved, and fertilization failure was observed even though calcium activation was given post ICSI. Proband, Mr. XX has been evaluated for pathogenic variations and polymorphism.

RESULTS

MALE FERTILITY SINGLE NUCLEOTIDE VARIATION ANALYSIS

Pathogenic variant related to the given phenotype was identified

List of significant variant identified related to the given phenotype:

Gene	Region	Variant*	Allele Status	Disease	Classification*	Inheritance pattern
<i>IQCN</i> (-)	Exon 3	c.2453delA (p.Gln818ArgfsTer56)	Homozygous	Spermatogenic failure 78 (OMIM#620170)	Pathogenic	Autosomal Recessive

*Genetic test results are based on the recommendation of American college of Medical Genetics [1-5].
No other variant that warrants to be reported for the given clinical indication was identified.

MALE POLYMORPHISM ANALYSIS

List of polymorphic variants (SNP) identified:

Gene Name	RsID	Variant	Genotype	Zygosity	Observed Phenotype	Recommendations
NOS3	rs1799983	c.894T>G	GG	Homozygous Polymorphism	Significant reduction in eNOS activity, leading to decreased nitric oxide bioavailability. This may contribute to impaired sperm function and an increased risk of male subfertility [6].	Minimize ROS exposure and handling time in vitro. L-arginine supplementation combined with antioxidants (CoQ10, vitamins C and E) might help improve semen parameters.
SOD2	rs4880	c.47T>C	TC	Heterozygous Polymorphism	Intermediate MnSOD activity. Increased ROS levels in semen. Increased sperm DNA fragmentation [7-8].	Clinical Correlation is strongly recommended.

Variant Interpretation

Interpretation of significant variant identified in Mr. XX related to the given phenotype

***IQCN*: c.2453delA**

Variant summary: A homozygous single base pair deletion in exon 3 of the *IQCN* gene (chr19:g.18265087delT, NM_001145304.2,Depth:145x) that results in a frameshift and premature truncation of the protein 56 amino acids downstream to codon 818 (p.Gln818ArgfsTer56) was detected. This variant is a frameshift variant which occurs in an exon of *IQCN* upstream of where nonsense mediated decay is predicted to occur. This variant is predicted to cause loss of normal protein function through protein truncation.

Population frequency: This variant has a minor allele frequency of 0.0055% in gnomAD database and has not been observed in 1000 genomes database.

Clinical & Literature Evidence: This variant has been classified as likely pathogenic in ClinVar database [3]. A frameshift variant in the same amino acid position (NM_001145304.2; c.2453_2454delAG; p.Gln818ArgfsTer9) has been previously reported in patient affected with total fertilization failure and male infertility in homozygous state [4].

In-silico predictions: The *in-silico* predictions of the variant is deleterious by MutationTaster2.

OMIM phenotype: Spermatogenic failure 78 (OMIM#620170) is caused by homozygous or compound heterozygous mutation in the *IQCN* gene (OMIM*620160). Spermatogenic failure-78 is an autosomal recessive disorder characterized by male infertility caused by an abnormal acrosome structure resulting from a manchette assembly defect. The acrosome is detached from the nuclear envelope of affected sperm, resulting in fertilization failure. In addition, defects in sperm flagellar assembly have been observed. This disease follows autosomal recessive pattern of inheritance [2].

Variant classification: Based on the evidence, this variant is classified as a pathogenic. **In this view, clinical correlation and familial segregation analysis are strongly recommended to establish the significance of the finding. If the results do not correlate, additional testing may be considered based on the phenotype observed.**

Recommendations

- Sequencing the variant(s) in the partner and the other affected and unaffected members of the family is recommended to confirm the significance.
- Genetic counseling is recommended.

Methodology: Single Nucleotide Variation - Single Nucleotide Polymorphism

SNV analysis: DNA extracted from the blood, was used to perform targeted gene capture using a custom capture kit (for SNV analysis – 1109 genes). The targeted libraries were sequenced to a targeted depth of 80 to 100X using SURFSeq 5000 sequencing platform. This kit has deep exonic coverage of all the coding regions including the difficult to cover regions. The sequences obtained are aligned to human reference genome (GRCh38.p13) using Sentieon aligner and analyzed using Sentieon for removing duplicates, recalibration and re-alignment of indels. Sentieon DNAscope has been used to call the variants. Detected variants were annotated and filtered using the VarSeq software with the workflow implementing the ACMG guidelines for variant classification. The variants were annotated using 1000 genomes (V2), gnomAD (v4.1.0,3.1.2,2.1.1), ClinVar, OMIM, dbSNP, NCBI RefSeq Genes. *In-silico* predictions of the

variant was carried out using VS-SIFT, VS-PolyPhen2, PhyloP, GERP++, GeneSplicer, MaxEntScan, NNSplice, PWM Splice Predictor. Only non-synonymous and splice site variants found in the coding regions were used for clinical interpretation. Silent variations that do not result in any change in amino acid in the coding region are not reported.

SNP analysis: Variant analysis (for SNP analysis – 115 genes) and interpretation is done using VarSeq Software. Extensive scientific literature, Information from variant analysis and disease specific databases, population specific research is used to interpret and recommend. All results are finally approved by medical geneticists.

Sequence data attributes

Total reads generated	11.52 Gb
Data ≥ Q30	96.78 %

Genetic test results are reported based on the recommendations of American College of Medical Genetics [1], as described below:

Classification	Interpretation
Pathogenic	A disease-causing variation in a gene which can explain the patients’ symptoms has been detected. This usually means that a suspected disorder for which testing had been requested has been confirmed
Likely Pathogenic	A variant which is very likely to contribute to the development of disease however, the scientific evidence is currently insufficient to prove this conclusively. Additional evidence is expected to confirm this assertion of pathogenicity.
Variant of Uncertain Significance	A variant has been detected, but it is difficult to classify it as either pathogenic (disease causing) or benign (non- disease causing) based on current available scientific evidence. Further testing of the patient or family members as recommended by your clinician may be needed. It is probable that their significance can be assessed only with time, subject to availability of scientific evidence.

Disclaimer

- The classification of variants of unknown significance can change over time. Anderson Diagnostics and Labs cannot be held responsible for it.
- Intronic variants, UTR, Promoter region variants and CNV are not assessed using this assay.
- Certain genes may not be covered completely, and few mutations could be missed. Variants not detected by this assay may impact the phenotype.
- The variations have not been validated by Sanger sequencing.
- The above findings and result interpretation was done based on the clinical indication provided at the time of reporting.
- It is also possible that a pathogenic variant is present in a gene that was not selected for analysis and/or interpretation in cases where insufficient phenotypic information is available.
- Genes with pseudogenes, paralog genes and genes with low complexity may have decreased sensitivity and specificity of variant detection and interpretation due to inability of the data and analysis tools to unambiguously determine the origin of the sequence data in such regions.
- In addition, this test does not cover detection of such structural variants in the genes *SMN1*, *SMN2*, *HBA1*, *HBA2*, *PMS2*, *GBA* and *CYP21A2* which have very high homology. Moreover, the ORF15 transcript isoform (RefSeq id: NM_001034853.1) of the *RPGR* gene associated with retinitis pigmentosa is not sufficiently covered in the test.
- Incidental or secondary findings that meet the ACMG guidelines can be given upon request [5].
- Result interpretation was done based on the literature evidence available at the time of reporting. The clinical significance of the polymorphic variants tested can change over time and Anderson Diagnostics & Labs cannot be held responsible for this.
- This is not a diagnostic test and so not to be considered as diagnosis of any disease. This test is meant only for understanding the polymorphism at a given position and its association with various clinical parameters.

References

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This report has been reviewed and approved by:



Sivasankar S, Ph.D
Molecular



Muthukumaran S, Ph.D
Clinical Bioinformatician



Sachin D. Honguntikar, Ph.D
Molecular Geneticist



Dr. G. Suriyakumar
Director



Fiona D'Souza, Ph. D
Head -Scientific Operations